

Project work — radians, arc length and sector area

Cambridge insert: the angle measure that makes every formula in calculus simpler, built and used in two lessons.

LESSON 77–78

2 × 40 MIN

ALGEBRA 10, §4.3 (EXTENSION)

P1 · 4.1–4.3

LESSON CLOCK

14'

18'

24'

20'

24'

8'

What a radian is	14 min
Converting	18 min
Arc and sector	24 min
The segment	20 min
The project	24 min
Homework	8 min

LEARNING OBJECTIVES

- Define the radian and convert between radians and degrees.
- Use $s = r\theta$ for arc length and $A = \frac{1}{2}r^2\theta$ for sector area.
- Find the area of a segment.
- Present a measured real object using radian formulas.

TEXTBOOK

National	pp. 337–342
Cambridge	Pure Mathematics 1, pp. 82–95
Workbook	P1 Exercise 4A–4C

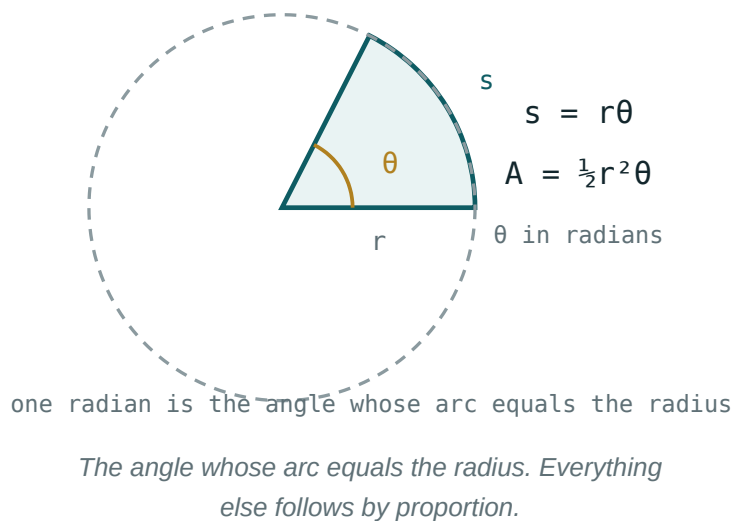
01

Explanation

What a radian is

DEFINITION

One **radian** is the angle at the centre of a circle subtended by an arc equal in length to the radius. It is a pure ratio $\frac{\text{arc}}{\text{radius}}$, so it has no units.



The whole circle has circumference $2\pi r$, so a full turn is 2π radians:

$$2\pi \text{ radians} = 360^\circ \quad \pi \text{ radians} = 180^\circ \quad 1 \text{ rad} \approx 57.3^\circ$$

Why bother. Degrees are an arbitrary Babylonian choice. Radians make the formulas below as simple as possible — and in Grade 11 they make $(\sin x)' = \cos x$ true, which it is not in degrees.

Converting

$$\text{degrees} \rightarrow \text{radians: multiply by } \frac{\pi}{180} \quad \text{radians} \rightarrow \text{degrees: multiply by } \frac{180}{\pi}$$

DEGREES	30°	45°	60°	90°	180°	270°
Radians	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$

SET THE CALCULATOR

Every wrong answer in this topic starts with the calculator in the wrong mode. $\sin 1$ is 0.0175 in degrees and 0.8415 in radians. Check the mode before every calculation.

Arc length and sector area

$$s = r\theta \quad A = \frac{1}{2}r^2\theta \quad (\theta \text{ in radians})$$

Both come from proportion: the sector is the fraction $\frac{\theta}{2\pi}$ of the whole circle, so its arc is that fraction of $2\pi r$ and its area that fraction of πr^2 .

IN DEGREES	IN RADIANS
$s = \frac{\theta}{360} \cdot 2\pi r$	$s = r\theta$
$A = \frac{\theta}{360} \cdot \pi r^2$	$A = \frac{1}{2} r^2 \theta$

The radian versions have no π and no 360 in them. That is the whole practical argument for radians.

The segment, and the project

A **segment** is the region between a chord and its arc — the sector minus the triangle:

$$A_{\text{segment}} = \frac{1}{2} r^2 \theta - \frac{1}{2} r^2 \sin \theta = \frac{1}{2} r^2 (\theta - \sin \theta)$$

THE PROJECT – TWO LESSONS, IN PAIRS

Find a real circular object or arc: a plate, a clock face, a fan, an arch of a madrasa, a wheel. Measure its radius and one angle. Then compute and present: the arc length, the sector area, the segment area, and one quantity of your own choosing.

One A3 sheet: a photograph or sketch with the measurements marked, the four calculations in radians, and one sentence on the accuracy of your measurement (Grade 11 will call these bounds).

Two pairs present in the last twenty minutes. The class asks one question each:

“How much would your answer change if your radius were 2 mm out?”

02 Worked examples

EXAMPLE 1 Convert 150° to radians and $\frac{5\pi}{6}$ to degrees.

$$(1) \quad 150 \times \frac{\pi}{180} = \frac{5\pi}{6}$$

$$(2) \quad \frac{5\pi}{6} \times \frac{180}{\pi} = 150^\circ$$

ANSWER

$$\frac{5\pi}{6} \text{ and } 150^\circ$$

EXAMPLE 2 A sector has radius 9 cm and angle 1.2 radians. Find its arc length and area.

$$(1) \quad s = 9 \times 1.2 = 10.8 \text{ cm.}$$

$$(2) \quad A = \frac{1}{2} \times 81 \times 1.2$$

$$(3) \quad = 48.6 \text{ cm}^2.$$

ANSWER

$$10.8 \text{ cm; } 48.6 \text{ cm}^2$$

EXAMPLE 3 Find the area of the segment cut off by a chord subtending $\frac{\pi}{3}$ at the centre of a circle of radius 6.

① Sector $\frac{1}{2} \times 36 \times \frac{\pi}{3} = 6\pi \approx 18.85$.

② Triangle $\frac{1}{2} \times 36 \times \sin \frac{\pi}{3} = 18 \times 0.866 \approx 15.59$.

③ Difference.

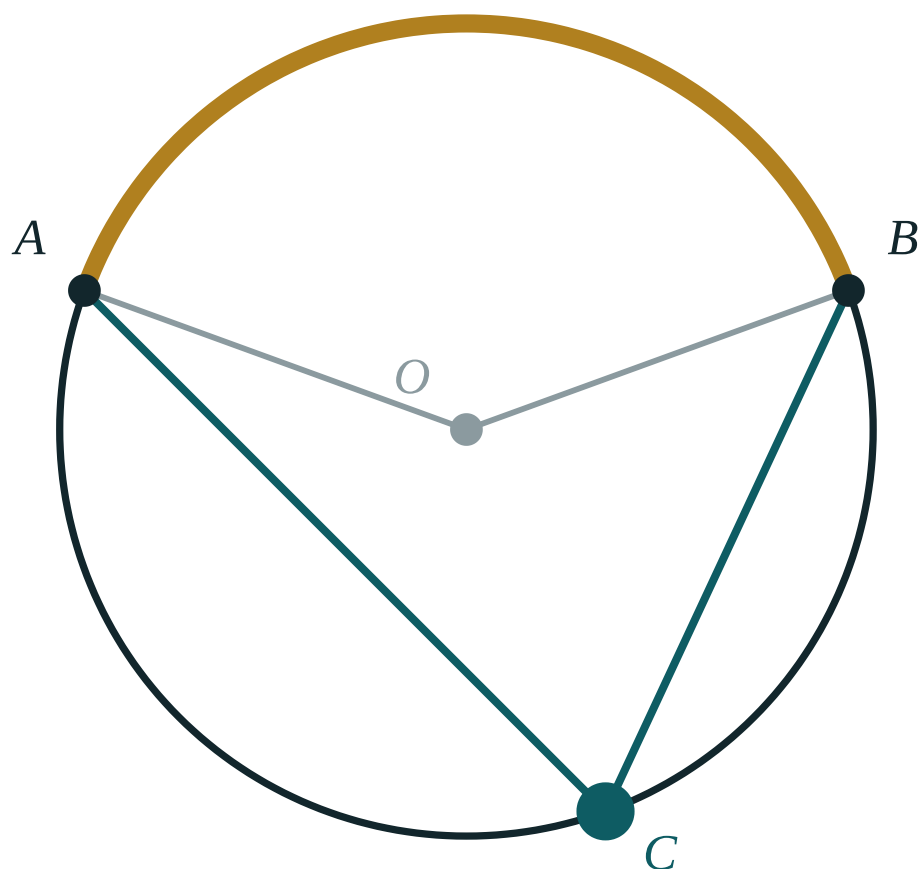
ANSWER

≈ 3.26 square units

03 Interactive model

Wrap a string of length r round the rim and mark the angle it subtends — that is one radian.

Arc, sector and segment



central $\angle AOB$ **140°**

inscribed $\angle ACB$ **70°**

ratio **2**

arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Radian	Radian	Радиян
Arc length	Yoy uzunligi	Длина дуги
Sector	Sektor	Сектор
Segment	Segment	Сегмент
Central angle	Markaziy burchak	Центральный угол
Conversion	O'tkazish	Перевод
Chord	Vatar	Хорда
Subtend	Qarshi turmoq	Стягивать
Angular measure	Burchak o'lchovi	Угловая мера

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

π radians equals:

90°

180°

270°

360°

60° in radians is:

$\frac{\pi}{6}$

$\frac{\pi}{4}$

$\frac{\pi}{3}$

$\frac{\pi}{2}$

Arc length in radians is:

$$2\pi r$$

$$r\theta$$

$$\frac{1}{2}r^2\theta$$

$$\frac{\theta}{360}2\pi r$$

Sector area in radians is:

$$\pi r^2$$

$$r\theta$$

$$\frac{1}{2}r^2\theta$$

$$r^2\theta$$

A segment's area is:

sector plus triangle

sector minus triangle

half the sector

the whole circle

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Convert 90° to radians

ANSWER $\frac{\pi}{2}$

2. Convert 45° to radians

ANSWER $\frac{\pi}{4}$

3. Convert π to degrees

ANSWER 180°

4. Convert $\frac{\pi}{6}$ to degrees

ANSWER 30°

5. Arc length, $r = 5$, $\theta = 2$

ANSWER 10

6. Sector area, $r = 4$, $\theta = 1$

ANSWER 8

7. A full turn in radians

ANSWER 2π

Medium · the standard question

7 PROBLEMS

1. Convert 150° to radians

ANSWER $\frac{5\pi}{6}$

2. Convert $\frac{3\pi}{4}$ to degrees

ANSWER 135°

3. Arc length, $r = 9$, $\theta = 1.2$

ANSWER 10.8

4. Sector area, $r = 9$, $\theta = 1.2$

ANSWER 48.6

5. A sector of arc 15 and radius 6: its angle

ANSWER 2.5 rad

6. A sector of area 24 and radius 4: its angle

ANSWER 3 rad

7. Perimeter of a sector, $r = 5$, $\theta = 1.4$

ANSWER 17

Hard · stretch and reason

7 PROBLEMS

1. Segment area, $r = 6$, $\theta = \frac{\pi}{3}$

ANSWER ≈ 3.26

2. Segment area, $r = 10$, $\theta = 2$

ANSWER ≈ 54.5

3. A sector of perimeter 20 and radius 6: its area

ANSWER 24

4. A sector of area 50 and arc 20: find r and θ

ANSWER $r = 5$, $\theta = 4$

5. A chord of length 8 in a circle of radius 5: the segment area

ANSWER ≈ 7.9

6. A clock's minute hand is 12 cm. Distance the tip travels in 20 minutes

ANSWER $8\pi \approx 25.1$ cm

7. Two circles radius 5 with centres 6 apart: area of overlap

ANSWER ≈ 30.4

07 Homework

Homework — 5 tasks

Bring the finished project sheet to the first lesson of Quarter IV.

- Convert 120° , 225° and 40° to radians; convert $\frac{7\pi}{6}$ and 2.5 rad to degrees.
- A sector has radius 12 cm and angle 1.5 rad. Find its arc length, area and perimeter.
- Find the area of the segment cut off by a chord subtending $\frac{2\pi}{3}$ in a circle of radius 8.
- A sector of area 60 cm^2 has radius 10 cm. Find its angle in radians and in degrees.

5. Complete the project sheet: the object, the measurements, the four calculations and the accuracy sentence.